Robustness of LLM Preference Alignment to Noisy Human Feedback

Direct Preference Optimization under Label Noise

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Problem and Hypothesis

Why this matters

- Modern alignment methods learn from comparisons: humans choose which answer is better.
- But real feedback is imperfect: annotators disagree, misclick, or prefer superficial traits like verbosity.
- If noisy labels are treated as truth, the model can learn the wrong preference signal.
- We test how quickly this failure appears as label noise increases.

Main hypothesis

More label noise should hurt vanilla DPO. Robust variants should degrade less.

Label noise model

$$\begin{cases} (y_c, y_r) & \text{with probability } 1-p \\ (y_r, y_c) & \text{with probability } p \end{cases}$$

Noise	Interpretation
0%	clean baseline
10%	mild noise
20%	realistic/moderate noise
30%	high noise
40%	severe noise

DPO and Variants We Compare

DPO optimizes the policy using the log-ratio margin

$$\delta = (\log p_\theta(y_c) - \log p_\theta(y_r)) - (\log p_{\mathrm{ref}}(y_c) - \log p_{\mathrm{ref}}(y_r)).$$

When labels are flipped, DPO's gradient weight *increases* for the flipped pair, actively reinforcing the wrong response.

Method	Loss function	Limitation
Vanilla DPO	$-\log \sigma(\beta\delta)$	Bias grows linearly with ϵ; gradient amplifies flipped updates.
cDPO	$-(1-\epsilon)\log \sigma(\beta\delta) - \epsilon \log \sigma(-\beta\delta)$	Biased estimator; noise-tolerant only at ϵ=0.5.
rDPO	$\dfrac{-(1-\epsilon)\log \sigma(\beta\delta) + \epsilon \log \sigma(-\beta\delta)}{1-2\epsilon}$	Requires knowing true flip rate ϵ; degrades if mis-specified.
ROPO	$\dfrac{4\alpha}{(1+\alpha)^2}[-\log \sigma(\beta\delta) + \alpha\, \sigma(-\beta\delta)]$	Needs tuning α

Experimental Setup

Training configuration

Model	Qwen2.5-7B-Instruct
Dataset	Anthropic HH-RLHF
Train examples	3,000
Clean eval examples	400
Training steps	8,000
Optimizer	AdamW
Quantization	4-bit NF4
Adapter	LoRA, rank 16
Learning rate	5×10^{-5}
β	0.1

Evaluation metrics

Clean preference accuracy

Fraction of clean eval pairs where the trained model scores y_c above y_r.

Clean preference margin

$(\log p_\theta y_c - \log p_\theta y_r) - (\log p_{\mathrm{ref}} y_c - \log p_{\mathrm{ref}} y_r)$ on clean eval pairs.

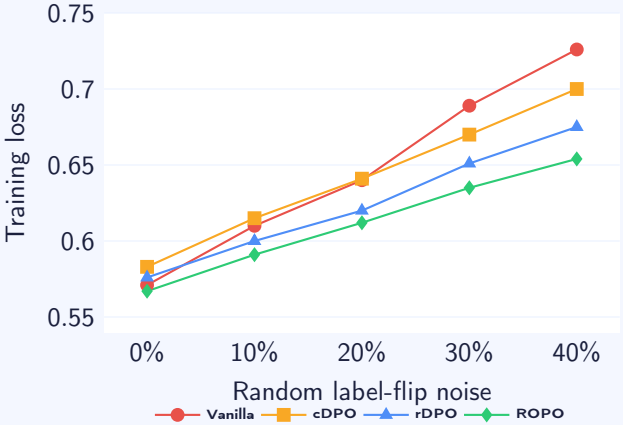
Training loss

Checks whether noise makes DPO optimization harder.

LLM-as-a-judge win rate

GPT-4o judges how often trained-model answers beat base-model answers.

Result 1: Training Loss Increases with Noise



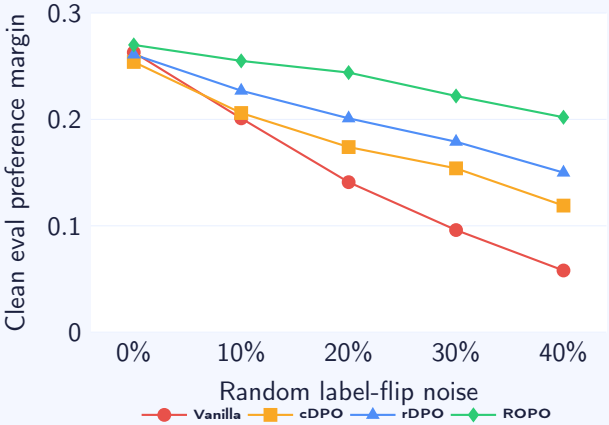
Interpretation

- Higher noise makes optimization harder.
- Vanilla DPO loss rises the most.
- ROPO keeps the lowest loss at every noise level.

Takeaway

Noise corrupts the training signal; robust losses absorb more of the damage.

Result 2: Noise Reduces Clean Preference Margin



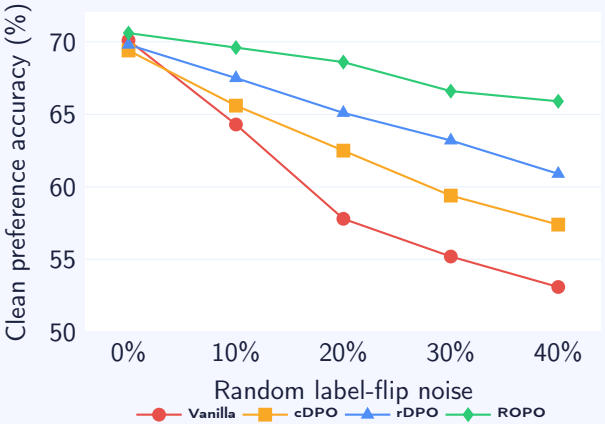
Interpretation

- Vanilla DPO loses most of its clean preference margin.
- rDPO and cDPO reduce the drop.
- ROPO preserves the largest margin at every noise level.

Takeaway

At 40% noise, ROPO's margin is 0.202 vs. vanilla's 0.058.

Result 3: Clean Preference Accuracy Drops



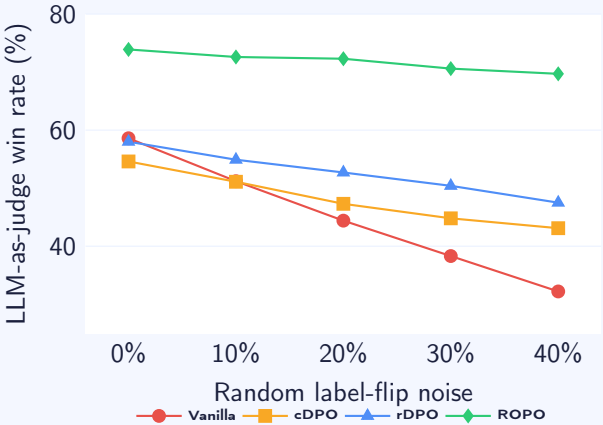
Interpretation

- Vanilla falls close to chance at 40% noise.
- Robust methods preserve more clean-pair ranking accuracy.
- ROPO remains best at every noise level.

Takeaway

At 40% noise, ROPO is 12.8 points above vanilla on clean accuracy.

Result 4: GPT-4o Judge Prefers Robust Models



Interpretation

- GPT-4o compares trained answers against base-model answers.
- Vanilla drops below the base model under high noise.
- ROPO stays strong even at 40% noise.

> **Takeaway**
>
> Robustness helps generated answer quality, not just fixed-pair ranking.

Conclusion

What we found

- Noisy preference labels substantially hurt vanilla DPO.
- At 40% random flips, vanilla drops to 53.1% clean preference accuracy and 32.2% GPT-4o judge win rate.
- Robust objectives reduce this degradation.
- ROPO is strongest overall: 65.9% clean preference accuracy and 69.7% judge win rate at 40% noise.

Final message

Main claim

Preference alignment is sensitive to noisy human feedback.

Practical implication

Noise-aware objectives preserve more clean preference signal and produce better judged responses.

Key references: Rafailov et al. 2023; Chowdhury et al. 2024; Liang et al. 2025.